

Finite Quantum Machine Learning for MEG Source Localization

A beamformer-seeded model of epileptic seizures, interictal spikes and ictal onset

Abstract. Epileptic source localization is an inverse problem in which a small number of neural generators must be inferred from noisy, time-resolved MEG measurements. We describe a finite hybrid workflow: a $10 \times 10 \times 10$ candidate source lattice is scored by a linearly constrained minimum-variance (LCMV) prior, and a compact variational quantum model refines the highest-scoring candidates. The formulation explicitly separates interictal spikes from seizure-period activity using finite-window statistics. All sums, state dimensions and optimization steps are bounded, making the method reproducible as a simulation protocol rather than an assertion of clinical performance.

1. Finite MEG forward model

Let $M = 64$ sensors, $G = 1000$ candidate locations and $T = 200$ time samples. The simulated sensor array is represented by y in $\mathbb{R}^{(M \times T)}$, with a finite set of dipole amplitudes x in $\mathbb{R}^{(G \times T)}$. For each sensor m , source g and sample t :

$$y_{m,t} = \sum_{g=1}^G L_{m,g} x_{g,t} + n_{m,t} \quad m = 1, \dots, M; \quad t = 1, \dots, T.$$

The lead-field coefficient is a bounded inverse-square surrogate, $L_{m,g} = (\|s_m - r_g\|^2 + 10^{-3})^{-1}$, where s_m is a sensor position and r_g is a lattice point. The additive term $n_{m,t}$ represents measurement noise; it prevents the inverse problem from becoming artificially exact.

2. Spike and seizure-window features

A transient interictal spike is represented on the normalized interval t in $[0,1]$ by a Gaussian-windowed oscillation. The finite sample sequence uses $t_k = k/(T-1)$:

$$q_k = \exp[-(t_k - \tau)^2/\sigma^2] \sin(2\pi f t_k), \quad k = 0, \dots, T-1.$$

For a candidate source g , define a finite spike-energy score over a window W_s and a seizure-energy score over W_z . The contrast is a dimensionless feature used by the localization objective:

$$E_g^{(s)} = \sum_{k \in W_s} z_{g,k}^2, \quad E_g^{(z)} = \sum_{k \in W_z} z_{g,k}^2, \quad R_g = E_g^{(z)} / (E_g^{(s)} + 10^{-6}).$$

Here $z_{g,k}$ is the reconstructed activity at lattice point g . A spike detector can be stated as the finite threshold rule $I_k = 1$ when $|q_k|$ exceeds the median absolute deviation threshold, and $I_k = 0$ otherwise. This separates an observable transient from the broader seizure window without claiming that one waveform defines every seizure type.

3. LCMV initialization

The sample covariance C in $\mathbb{R}^{(M \times M)}$ is regularized by λI , with $\lambda > 0$. For each finite lattice point g , the scalar LCMV power is:

$$P_g = [L_g^T (C + \lambda I)^{-1} L_g]^{-1}, \quad g = 1, \dots, G; \quad g_0 = \arg \max_g P_g.$$

The index g_0 seeds the QML state rather than replacing the complete search. In the associated simulation, the prior map is normalized by $P_{\blacksquare} = \max_g P_g$ and the initial candidate is retained as an auditable baseline.

4. Finite variational QML refinement

Encode a local neighborhood of $K = 8$ candidate locations into three qubits. A parameterized circuit $U(\theta)$ produces a finite probability distribution $p_j(\theta)$ over $j = 0, \dots, K-1$. The source loss combines data mismatch, localization distance

and a seizure-feature penalty:

$$J(\theta) = \alpha \sum_{k=0}^{T-1} [y_k(\theta) - y_k]^2 + \beta \sum_{j=0}^{K-1} p_j(\theta) d_j^2 + \gamma [1 - R_{j^*}], \quad j^* = \arg \max_j p_j(\theta).$$

With step size eta and Q = 12 bounded optimization steps, the parameter update is $\theta^{(u+1)} = \theta^{(u)} - \eta \nabla J(\theta^{(u)})$, for u = 0,...,Q-1. Gradients may be estimated by the parameter-shift identity for a gate parameter θ_i :

$$\partial J / \partial \theta_i = [J(\theta_i + \pi/2) - J(\theta_i - \pi/2)] / 2.$$

The finite stopping rule is $\min\{u: |J(\theta^{(u)}) - J(\theta^{(u-1)})| < 10^{-4}\}$, with a hard limit of Q iterations. This makes the computational claim testable: the output is a candidate source distribution and a spike/seizure feature vector, not a clinical diagnosis.

5. Epileptic seizure prediction

Prediction is defined here as a finite warning score derived from the localized QML activity. For each sample k, let a_k be the normalized source activity and h_k the finite feature vector containing spike energy, seizure-window energy and spectral contrast. A logistic readout gives:

$$p_k = [1 + \exp(-(\omega_0 + \omega^T h_k + \omega_a a_k))]^{-1}, \quad d_k = 1[p_k \geq 0.70].$$

The example uses a 0.70 decision threshold only to illustrate the finite rule. The characteristics of the two modeled phenomena are: an interictal spike is brief, high-slope and localized in time; ictal activity is sustained, has elevated window energy and produces a persistent probability elevation. These are simulation features, not diagnostic criteria.

Characteristic	Spike-like activity	Seizure-like activity
Duration	Brief Gaussian-windowed transient	Sustained elevated envelope
Energy	Concentrated in W_s	Elevated across W_z
Prediction signal	Transient feature peak	p_k remains above threshold

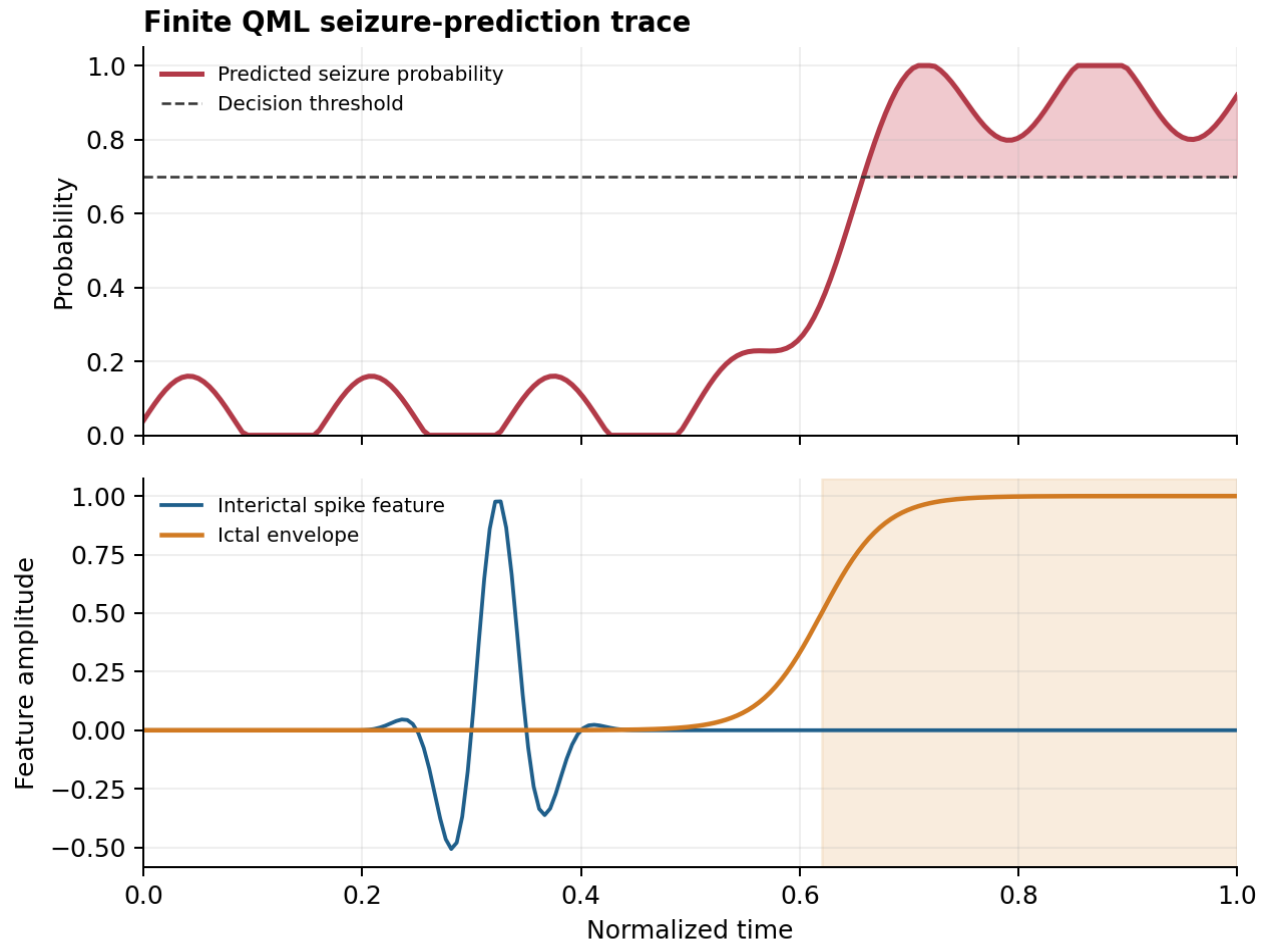


Figure 1. Deterministic finite example. The upper panel shows the predicted probability and thresholded seizure decision. The lower panel contrasts a brief interictal spike feature with a sustained ictal envelope; the shaded region marks the modeled seizure window.

6. Reproducible simulation protocol

Quantity	Finite setting	Purpose
Sensors M	64	Noisy MEG observations
Source grid G	$10 \times 10 \times 10 = 1000$	Candidate locations
Samples T	200	Transient waveform window
QML neighborhood K	8 states / 3 qubits	Local refinement
Optimization steps Q	12 maximum	Bounded parameter updates

Limitations. The equations define a finite computational model and do not establish diagnostic accuracy, seizure onset-zone sensitivity, or patient-level safety. Real deployment would require calibrated head models, artifact rejection, clinical annotations, prospective validation and appropriate regulatory review.

7. Conclusion

A beamformer-seeded QML search provides a compact way to rank candidate generators while preserving explicit finite bounds on sensors, locations, samples and optimization steps. The same objective can report both a localized source estimate and interpretable spike-versus-seizure window features, providing a clear bridge for future MEG/MRI

validation studies.